CODE AXIS HACKATHON | ROUND 1

THE PLAN

BrandLens AI

An explainable, human-in-the-loop platform
for pre-publication brand governance

“Publish with confidence. On-brand, on-fact, on-record.”

SOURCE	REVIEW	DECISION
Approved brand knowledge	Explainable checks	Human approval

TEAM QUANTUM CODERS

Siyaram Meena • Vedant Raju Bhagat • Mayank Gupta • Shreya Singh

Problem statement provided by Inkloom AI

STATUS — ROUND 1 PLAN No prototype or code has been built. All product capabilities, architecture and safeguards described here are proposed. No team interviews or surveys have been completed.

Prepared for the Round 1 PDF submission. Working product name and tagline; subject to team confirmation.

Executive summary

The challenge. Organisations publish through employees, agencies, freelancers and AI tools. Content can include unsupported claims, outdated facts, inconsistent tone, missing disclosures or off-brand assets—and manual review can be late, fragmented and hard to audit. Inkloom AI asks teams to propose a functional, explainable pre-publication brand-governance product that checks against approved materials, suggests intent-preserving corrections and keeps people in control. [1]

Our proposal. BrandLens AI is a planned web application where a brand keeps approved guidelines, facts, claims and terminology in a private workspace. Creators join through BrandPIN (an access feature), submit content, and receive highlighted findings linked to supporting passages—or an explicit “knowledge gap” where evidence is absent. A human reviewer then requests changes, approves or rejects. The product is not intended to publish automatically or replace legal review.

Why this problem matters. ASCI’s FY2025–26 report says 98% of 9,841 ads scrutinised required some modification; this is a monitored corpus, not a prevalence estimate for every Indian ad. [2] McKinsey’s 2025 survey reports AI inaccuracy among commonly experienced AI risks. [4] These sources support the need for careful review, but do not prove BrandLens will improve outcomes.

At a glance

Item	Round 1 plan
Product	BrandLens AI (working name); pre-publication brand-governance assistant
Primary users	Creators, social-media teams, agency/freelance contributors; brand reviewers
Core approach	Approved-source retrieval + deterministic checks + LLM-assisted language review + human approval
Proposed stack	Next.js/React · FastAPI · PostgreSQL/pgvector · Gemini API (synthetic data only)
MVP focus	Text/document review, evidence-linked findings, advisory score, human approval workflow
Built so far	Nothing implemented; Round 1 is planning only

ROUND 1 CONTEXT The organiser confirms this round is a PDF plan only; code starts in Round 2. Round 2 and Round 3 details below are provisional because no later-round rules were provided.

1. Problem statement and our interpretation

Official challenge statement — condensed faithfully from the Inkloom AI brief supplied by the team. This is the organiser's requirement, not our proposed solution. [1]

Official requirement	What it asks teams to address
Problem	Multiple employees, agencies and AI tools make consistent, trustworthy brand identity difficult; incorrect claims, outdated information, inconsistent language and off-brand visuals can harm credibility.
Product outcome	Build an intelligent platform that checks content before publication against approved guidelines, facts and assets.
Detection	Address tone, terminology, messaging, visual consistency, accessibility and unsupported promotional claims across content formats.
Explanations and edits	Explain each issue and suggest corrections that preserve the creator's original purpose; do not simply reject.
Governance	Support creator/manager/authorised reviewer collaboration, comments, approvals, change tracking, repeat-error insights and unclear-guideline insights.
Principles	Protect confidential material and balance automation with human control, reliability and explainability.

Plain-language interpretation

A brand-aware pre-publication checker: compare a draft with the brand's own approved rules and facts; show the exact sentence, the relevant source and a safer edit; let an authorised person decide what happens next. It should say when the available evidence is insufficient rather than inventing a rule.

Formal interpretation for judges

We interpret the need as a governed content-review layer that retrieves approved brand knowledge, evaluates content against that evidence, returns auditable findings, supports minimal edits and a role-scoped human approval process, and aggregates repeat issues—without treating an AI score as a compliance verdict.

Problem analysis

Question	Our interpretation
Why does it matter?	Trust, rework, regulatory exposure and accessibility are at stake; published evidence supports the context, not product effectiveness.
Who is affected?	Creators and reviewers directly; organisations that adopt/fund the workflow; audiences who receive the published content.
Root causes	Guidance is scattered or hard to apply; many contributors work across campaigns; approved facts change; manual review bottlenecks; generic AI is not grounded in private approved claims.
Current methods and gaps	Manual email/chat review, shared style guides, brand portals and general AI tools. Methods can be slow or disconnected; capabilities vary and require independent product evaluation.
Constraints	Evidence-linked explanations, confidentiality, realistic formats, human final decision, secure role access and a working product by final round.

2. Is this a real problem? Evidence review

We conducted desk research using regulator, institutional, survey and technical sources. It establishes that relevant risks exist; it does not establish that our specific design works. The team has not conducted interviews, surveys or observations. Published figures are presented with their scope and limitations.

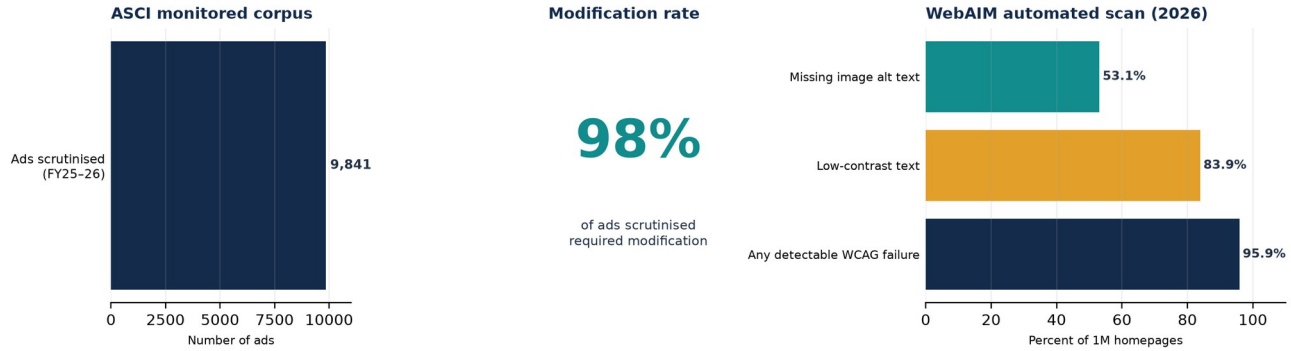


Figure 1. Two independent snapshots; scales and populations differ. ASCI is a monitored ad corpus; WebAIM is an automated homepage scan. [2,5]

Finding / period	Source	Relevance	What it does not prove
9,841 ads scrutinised; 98% required some modification (FY Apr 2025–Mar 2026).	ASCI annual report, 2026 [2]	Shows frequent issues among ads ASCI examined; 97.3% appeared digitally.	Not a random sample of all ads; not an estimate of all-market violation prevalence.
Indian ad guidelines define misleading-ad and endorsement expectations (2022).	CCPA official guidelines [3]	Supports a real governance context for claims/disclosures.	Does not mean BrandLens can give legal advice or guarantee legal compliance.
51% of AI-using organisations reported ≥1 negative consequence; inaccuracy was among most-cited risks (2025 survey).	McKinsey Global Survey [4]	Supports verifying AI-assisted content against trusted sources.	Global, self-reported, not marketing-specific.
95.9% of sampled homepages had detectable WCAG failures; 83.9% low contrast; 53.1% missing alt text (Feb 2026).	WebAIM Million [5]	Supports accessibility checks as an adjacent, partly automatable concern.	Automated checks only; homepages are not all marketing assets or all accessibility issues.
77% of surveyed companies reported dealing with off-brand content (2021).	Marq/Lucidpress survey [6]	Suggests a gap between brand intent and content practice.	Older vendor-sponsored, self-reported survey; not causal and not representative of all organisations.

What is supported vs. still to validate

Published evidence supports the existence of advertising review, AI accuracy, brand consistency and accessibility concerns. It does not show how much time our target users lose, whether they trust source-cited AI findings, or whether our system is accurate. Those require primary validation and a labelled evaluation set.

PLANNED VALIDATION — NOT COMPLETED In Round 2, interview 10–15 creators, social-media managers, freelance/agency contributors and reviewers. Ask about the last correction, guideline access, review time, repeat errors, trust requirements and what data cannot leave their organisation. Test a prototype only after it exists; report results only after collection.

3. Users, needs and current alternatives

Group	Role / need	How the proposal helps	Adoption barrier / trust condition
Creators, copywriters, social-media staff	Primary users; need fast clarity about approved claims and tone.	Shows relevant rule beside the draft; suggests minimal edits.	Extra step and AI distrust; must be quick, explainable and editable.
Agency / freelance creators	Primary users; work across brands with confidential rules.	BrandPIN request + admin-approved, role-scoped access.	Must avoid exposing another client's data; access must be revocable.
Managers / authorised reviewers	Daily reviewers and decision-makers; need backlog prioritisation and traceability.	Severity-ranked findings, comments, versions and final decision control.	Must be able to inspect evidence, override flags and maintain accountability.
Brand admins / marketing or compliance leads	Potential adopter/funder; need governance, approved knowledge and auditability.	Private workspace, controlled knowledge versions and repeat-issue analytics.	Accuracy, privacy, setup effort, cost and fit with existing workflow.
Consumers / public	Indirect beneficiaries.	Potentially fewer unsupported or inaccessible messages if adopted effectively.	Impact is hypothetical until validated in real use.

Users are the people doing review work; beneficiaries include audiences and brand reputation; adopters/funders are brands, agencies or organisations. They are distinct groups with different success criteria.

Illustrative personas — assumptions, not research participants

Persona	Job to be done	Pain point	Trust trigger
Creator at a small brand	Submit campaign copy on time.	Does not know which claim/disclaimer is allowed.	Exact source, clear explanation, rewrite preserving intent.
Brand manager	Approve consistent, defensible content.	Repeated corrections and limited review time.	Evidence, status history, override controls.
Freelance creator	Work for multiple clients safely.	Guidance differs; internal files may be restricted.	Admin-approved BrandPIN scope and explicit permissions.

Existing approaches (indicative public-product review)

Approach	Strength	Gap / caveat for our use case
Acrolinx / Markup AI [7]	Enterprise content guidance, style and terminology capabilities are marketed.	Public product information is not a test of every workflow; pricing/fit for small teams needs checking.
Frontify Brand Assistant [8]	Brand guidance and asset access; copy review against guidelines is described in help materials.	We have not tested the product; feature coverage and permissions depend on plan/configuration.
General-purpose LLM	Fast drafting and rewriting.	Without approved brand sources and governed review workflow, factual and brand fit still need human checks.
Manual email/chat review	Flexible human judgement.	May be fragmented and difficult to measure; effectiveness depends on team process.

4. Proposed solution and defensible value

One-line description: BrandLens AI is a proposed explainable reviewer that checks draft content against a brand's approved guidelines and facts before publication, then hands the decision to an authorised human.

Problem-to-solution mapping

User problem	Planned feature	How it helps	Measure to test
Rules are hard to find	Retrieve and cite relevant approved passages.	Makes the rule visible where the issue appears.	% findings with valid citation or "knowledge gap"; clarity rating.
Unsupported claims or missing disclaimers	Fact/claim retrieval + deterministic required-element checks.	Flags unsupported statements without asserting legal judgment.	Recall/precision on labelled synthetic examples.
Tone/terminology drift	LLM-assisted tone check + approved term list.	Highlights text and proposes minimal alternative.	Reviewer acceptance and precision.
Review bottleneck / no audit	Risk prioritisation, comments, approvals, version history.	Clarifies next action and who decided.	Review time and revision rounds vs baseline.
External creators need limited access	BrandPIN join request + admin-granted role scope.	Connects contributor without PIN itself unlocking brand data.	Access-control tests; zero cross-brand reads.

Illustrative advisory score

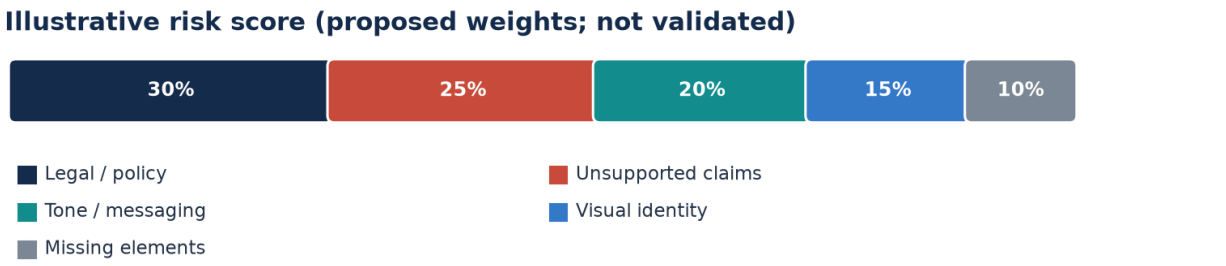


Figure 2. Initial scoring concept; weights are proposed, not calibrated or implemented.

Any score is advisory only. A critical single finding should not be hidden by an average, and an authorised reviewer always makes the final decision.

PRODUCT NAME BrandLens AI is a working name; BrandPIN is a feature of the product, not a separate product.

5. End-to-end workflow

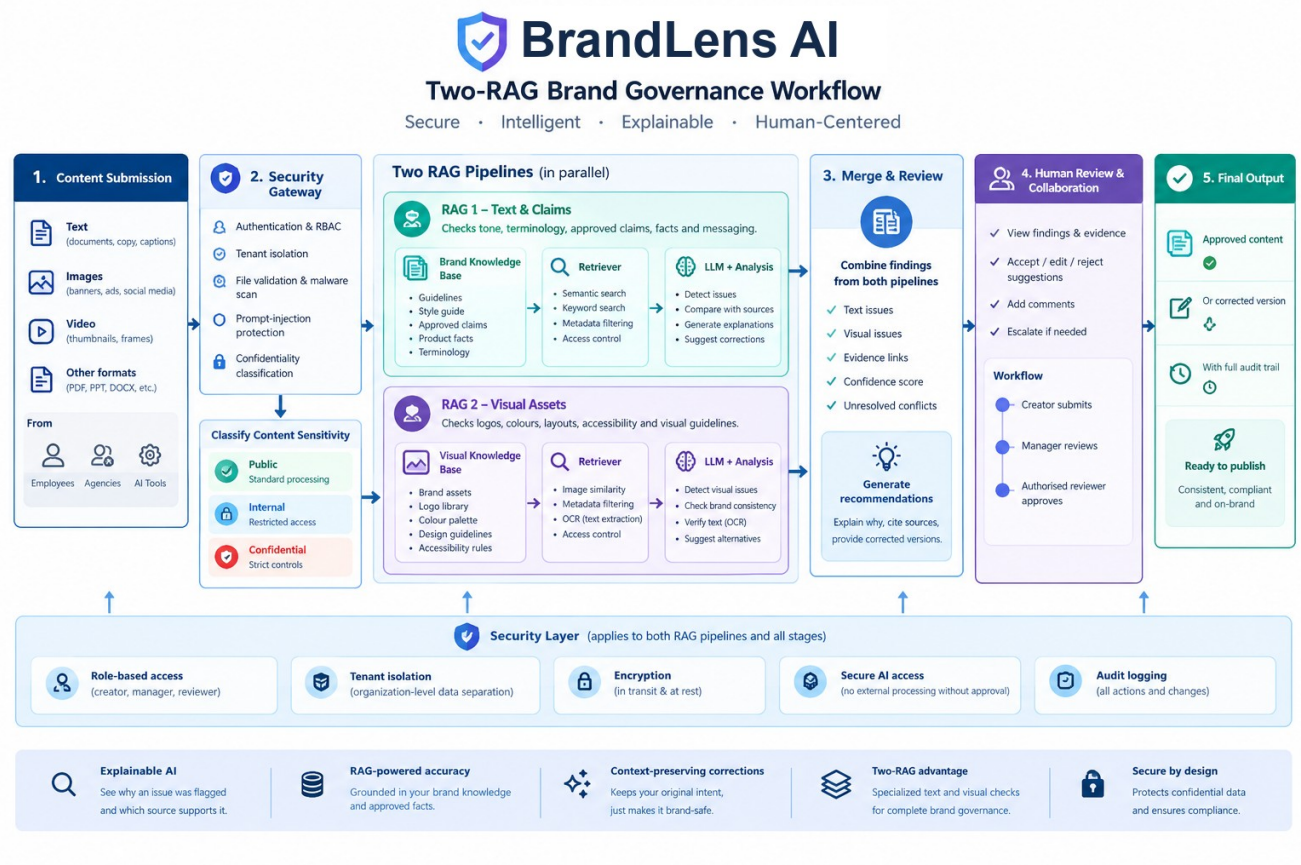


Figure 3. Supplied workflow visual adapted to the BrandLens AI concept; all features shown are proposed, not implemented.

Scope note: this is a target-state illustration. The planned MVP focuses on synthetic text/document review; visual or video analysis and publishing integrations are future scope. A human reviewer makes the final decision. The product does not guarantee legal compliance or publish automatically.

Step	User / system action	Safeguard / output
1. Set up	Brand admin creates workspace and uploads approved guidelines, facts, terminology, required disclosures and assets.	Version, category and brand metadata; admins own approved knowledge.
2. Join	Creator signs in, enters BrandPIN and requests access; admin grants role/campaign permissions.	Pending users see no private data; PIN expires/rotates and can be revoked.
3. Submit	Creator pastes text or uploads supported text document.	Validate type, size and empty content; unsupported input gets a clear message.
4. Retrieve & inspect	Split content into segments; fetch relevant approved sources; run objective rules and AI language checks.	Tenant filter; each finding requires source ID/quote or “knowledge gap”.
5. Report & act	Show highlighted issues, severity, advisory 0–100 score, minimal correction.	Low confidence or conflicting sources → “needs human review”; no auto-publish.
6. Review & learn	Creator revises; reviewer approves, requests changes or rejects; retain history and aggregate repeat issues.	Human decision and guideline updates; analytics do not silently change rules.

6. Technical data flow and architecture

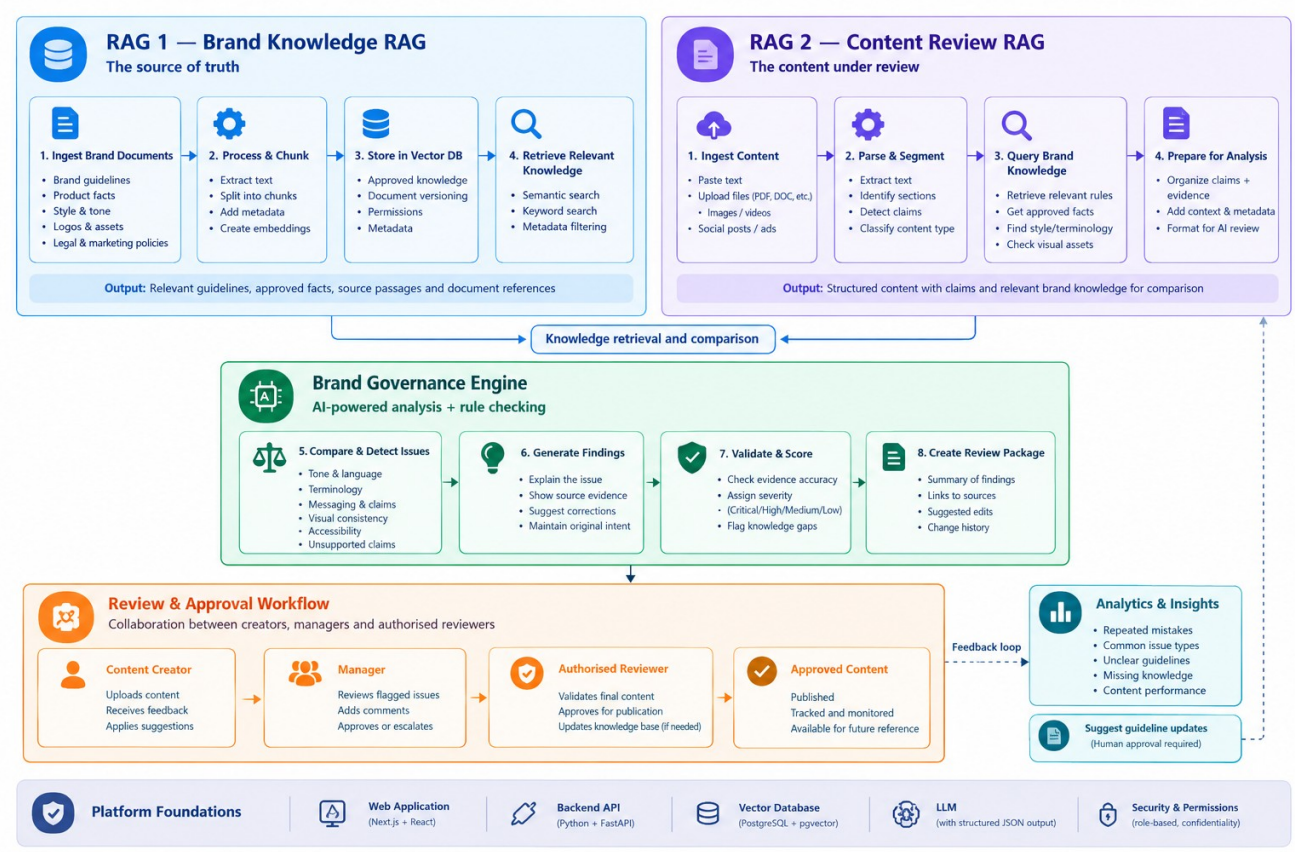


Figure 4. Supplied two-RAG workflow visual; interpreted as a proposed target-state, not a working implementation.

Scope note: the first build prioritises the text/document path and synthetic data. Retrieval can help ground findings but cannot guarantee correctness; human review remains essential. Visual checks, integrations and all security controls require implementation and testing.

Two retrieval lanes (RAG)

Brand Knowledge (RAG 1) stores only admin-approved brand sources with version, campaign and permission metadata. Content Review (RAG 2) parses the submission and asks for relevant passages from RAG 1; draft content is not treated as authority. The governance service combines retrieval with deterministic checks and constrained LLM output. Retrieval can improve grounding, but does not guarantee correctness; evaluation is required. [9]

Error and edge-case behaviour

Scenario	Planned response
LLM/API unavailable or quota hit	Retry briefly; then rules-only mode, prominent status banner, item remains in human queue; demo fallback cached synthetic examples.
No relevant approved source	Return “knowledge gap”; do not guess; notify admin to review or add approved guidance.
Malformed model output / nonexistent source ID	Reject output, retry once with schema repair, otherwise discard and log a minimal event.
Low-confidence or conflicting evidence	Mark “needs human review”; never auto-approve/reject.
Invalid file / unsupported language or format	Explain what could not be checked; do not imply clean/compliant; avoid storing invalid input.
Prompt injection within draft	Treat submission as data, not instructions; validate output against schema and retrieved source IDs.

7. MVP scope and technology choices

The MVP must demonstrate one complete path, not every format or enterprise capability. No component listed here is implemented yet.

Priority	Feature	User problem / demonstration	Complexity / success check
P0 essential	Workspace, authentication, roles and BrandPIN request/approval.	Scoped creator access; verify no unapproved or cross-brand data access.	Medium; role/isolation tests pass.
P0 essential	Upload approved PDF/DOCX/TXT + retrieve text passages.	Reviewer can find the relevant brand rule.	Medium; top-3 retrieval test on synthetic prompts.
P0 essential	Text submission + objective rule checks + LLM-assisted review.	Find tone/claims/required elements with cited evidence.	High; evaluate precision/recall on labelled set.
P0 essential	Highlighted findings, advisory score and human review status/history.	Creator can revise; reviewer can approve/request change/reject.	Medium; complete end-to-end demo.
P1 if time	Basic recurring-issue analytics and rewrite comparison.	Demonstrate feedback on repeated issues and preserve intent.	Low–medium; totals reconcile to stored findings.
Stretch / defer	OCR/image, logo/colour checks, multilingual support, SSO, browser/Canva integrations.	Broader formats and integrations.	Do not block core MVP; defer full visual checks.

Proposed stack

Component	Choice and purpose	Why / alternative / effort
Frontend	Next.js + React; responsive creator/reviewer UI.	Familiar web patterns and deployable app; Vite+React alternative; medium.
API / AI service	Python FastAPI; workflow and review endpoints.	Typed validation and Python AI ecosystem; Django REST alternative; medium.
Data	PostgreSQL + pgvector; structured workflow and embeddings.	One store for relational + vector MVP; managed Postgres alternative; medium.
Language model	Gemini API structured output for tone/claim assistance.	Replaceable provider boundary; JSON schema helps shape output but not validate truth. Free-tier terms/rates change; use synthetic content only. [10]
Embeddings / parsing	Open-source local sentence embeddings; pypdf/python-docx.	Avoid per-call embedding cost; test language quality and licenses; medium.
Hosting / testing	Vercel frontend + Render-like API + managed Postgres; GitHub; pytest/Playwright.	Free/low-cost tiers may have limits/sleep; use a local demo fallback; low–medium.

Data, APIs and fallbacks

Source / access	Use and constraints	Fallback / status
Team-created synthetic brand kits and 50–100 labelled sample posts. No external API or key.	Build two fictional brand workspaces with seeded violations; team will create and label examples. No real customer material; internal, to be created.	Planned dataset; pair-label a subset and resolve disagreements.
CCPA official guideline PDF [3]; ASCI annual report [2]. Public official documents; no API key.	Use as regulatory/context references for human-reviewed rule examples; verify applicable wording before encoding.	Use seeded synthetic rules if source interpretation is uncertain; not legal advice.
Gemini API [10,13]. API key/Google account required; quotas, model availability and pricing change.	Structured output for language review; send only minimum necessary context. Free tier may have different data-use terms; synthetic data only.	Rules-only mode + cached synthetic demo; provider interface is replaceable.

8. Security, privacy and responsible design

IMPLEMENTATION STATUS These are planned controls, not implemented safeguards or an audit result. Do not use real client content in the demo.

Area	MVP safeguard	Before real-world use
Minimise data	Synthetic brand kits/content; collect only data needed; document what is sent to LLM.	Consent, retention/deletion controls and data-processing review.
Identity and tenancy	Hashed passwords or managed auth; expiring sessions; role-based access; brand_id filters; deny by default.	Independent access-control review; stronger identity/SSO if needed.
BrandPIN	Random, rate-limited, expiring invitation; admin approves; scoped roles; revoke access.	Abuse monitoring and incident response.
Web/API security	HTTPS; server validation; safe file type/size limits; parameterised database access; output encoding.	Threat model, dependency scan and security assessment.
Secrets / providers	Keep API keys in environment secrets, not repository; synthetic data on free-tier services.	Review vendor retention/training terms; choose paid/self-hosted option before client data.
AI reliability	Source IDs and quotes, schema validation, “knowledge gap”, confidence handling and human decision.	Representative evaluation, drift monitoring and documented limitations.
Logging / retention	Log event IDs/status, not full draft text; provide basic delete for demo data.	Defined retention policy, deletion verification, backups and audit procedures.
Accessibility / fairness	Keyboard support, labels/contrast; test writing variations and Hindi/Hinglish where supported.	Assistive-technology testing and broader language/sector evaluation.

Project-specific risk register

Risk	Likelihood / impact	Mitigation	Owner
Wrong or fabricated AI flag	Medium / High	Require source; schema check; knowledge-gap path; reviewer can dismiss.	Shreya Singh
Missed serious issue	Medium / High	Deterministic rules where possible; labelled recall testing; no AI auto-approval.	Shreya Singh
Cross-brand information leak	Low / Critical	Tenant IDs, least privilege, access tests on every endpoint.	Vedant Raju Bhagat
API outage / quota during demo	Medium / High	Rules-only fallback; cached synthetic walkthrough; provider interface.	Siyaram Meena
Scope exceeds time / integration bottleneck	High / High	Freeze MVP, weekly cut line, mock API contract; drop visual checks first.	Siyaram Meena
Confidential draft sent externally	Medium / High	Synthetic data only during event; minimum context; do not use free tier with real data.	Siyaram Meena

9. Team allocation and three-round roadmap

Member / institution	Primary ownership	Additional / shared duties
Siyaram Meena — DTU Team Leader	Website lead, product coordination and overall delivery.	Largest coordination scope: own core website; define scope and architecture; integrate all modules; coordinate research and user validation; co-design score; review workflow/analytics; own release, QA, demo, pitch and submission documentation.
Vedant Raju Bhagat — DTU	Backend and database.	API/data model, auth/RBAC, BrandPIN endpoints, workflow states, versions/audit, tenant security, deployment and integration tests; pair on service integration.
Mayank Gupta — DTU	Frontend and UI/UX.	Design system, creator/reviewer screens, risk report/highlights, accessibility, API wiring, usability test and demo visuals; co-design review flow with lead.
Shreya Singh — Banasthali Vidyapith (BU)	FastAPI and AI.	Brand knowledge ingestion/retrieval, prompts/JSON schema, LLM integration/fallbacks, scoring implementation, synthetic labels and precision/recall evaluation; AI-safety tests.

RACI (R = Responsible, A = Accountable, C = Consulted, I = Informed)

Deliverable	Siyaram	Vedant	Mayank	Shreya
Scope, requirements, research plan	A/R	C	C	C
Website integration and delivery	A/R	C	R	C
API, DB, auth, BrandPIN, audit	A	R	C	C
UI/UX and frontend	A	C	R	C
RAG, LLM, review scoring	A	C	I	R
Security, deployment and integration tests	A	R	C	C
Evaluation, demo, documentation	A/R	C	C	R

Coordination: GitHub with protected main branch and peer-reviewed pull requests; a shared task board and API contract; two short syncs per week; weekly milestone demo; one documented decision log.

Three rounds

Round / status	Focus and deliverables	Gate / fallback
Round 1 — PDF plan	Problem reading, users, solution, workflow, feature/stack plan, team, impact and future scope.	Submission checklist complete; no code is required.
Round 2 — provisional	Build core journey: workspace → BrandPIN request → approved knowledge upload → text submission → cited findings/score → human decision. Synthetic kit, labelled test set, integrated demo and feedback plan.	Milestones: skeleton/API contract; retrieval; end-to-end test; evaluation. If late, drop analytics and image checks.
Round 3 — provisional	Stabilise, test edge cases/access controls, measure KPIs, fix highest-impact issues, rehearse and document limitations.	Feature freeze, regression pass, two rehearsals and recorded/local fallback if hosting fails.

Member-wise ownership across rounds

Member	Round 1 plan	Round 2 build	Round 3 finalise
Siyaram	Scope, brief interpretation, coordinate plan and submission.	Own web integration/review flow; coordinate research and demo; manage cut line.	Integration and bug triage; KPI/demo, pitch and final documentation.
Vedant	Review backend feasibility and security requirements.	Schema, auth/RBAC, BrandPIN, workflow history, deployment and access tests.	Security hardening, rate limits, backups and deployment stability.
Mayank	User journey and interface requirements.	Design system, submit/review UI, API wiring and usability checks.	Accessibility/UI polish, edge states and demo visuals.
Shreya Singh	RAG/AI feasibility and evaluation plan.	Ingestion/retrieval, structured findings, scoring and labelled evaluation.	Tune prompts/weights; edge-case testing and evaluation report.

10. Impact, roadmap and readiness

Expected impact and how to test it

Metric / intent	Definition and method	Proposed target or baseline
Source coverage	Finding has a valid approved source ID/quote or explicit knowledge-gap label; automated check.	100% of findings (design target).
High-severity recall	Seeded high-risk cases found / all seeded high-risk cases in labelled synthetic set.	≥0.80 as a proposed test target, not achieved.
Finding precision	Reviewer-accepted flags / all flags reviewed on labelled set.	≥0.70 proposed; report sample size and disagreements.
Review time / revisions	Minutes from submission to decision; count change requests in workflow.	Measure manual baseline first; propose improvement only after baseline.
Access safety	Cross-tenant read attempts succeeding in automated tests.	Zero successful reads in tests; not a security certification.
Usability	Participants completing the review flow without assistance.	Set a target before usability sessions; no sessions completed yet.

Expected impact and scalability

Hackathon: demonstrate an evidence-grounded review flow on synthetic content. Near term: pilot with a few student societies, local brands or agencies only after consent and security review. Longer term: fewer avoidable corrections and more traceable approvals are hypotheses, not proven outcomes. Scale via tenant-scoped data, indexed retrieval and replaceable model services; real deployment requires load tests, paid/contract-reviewed APIs, retention controls, security review and partnerships.

Post-event stages (aspirational)

0–3 months: evaluate with pilot teams and improve reliability. 3–9 months: consider browser/document-tool integrations and Hindi/Hinglish only after testing. 9–18 months: production security, privacy, deletion and monitoring. Longer term: broader formats and calibrated scoring research. Commercial model (workspace subscription for small teams or enterprise tier) is an assumption to test, not a commitment.

Round 1 requirement checklist

Requirement	Status	Notes
Problem reading; users; solution; workflow; intended features	Covered	Official Inkloom AI brief kept distinct from team interpretation.
Stack; team roles; three-round plan; impact; future direction	Covered	Round 2/3 explicitly provisional.
Team name and four participant details	Covered	Quantum Coders; names/institutions listed.
Prototype or code	Not required for Round 1	None exists; no implementation is claimed.
Primary user research	Not completed	Desk research is included; interviews/usability are planned for later.
Team leader email and Drive link	Needs action	Email required in submission form but not provided here; create link, enable Anyone with link → Viewer, test in incognito.

READINESS The PDF content is a complete Round 1 plan draft. Before submitting, the leader must enter their email in the organiser form, upload this PDF to Drive, set Anyone with the link → Viewer and verify access in an incognito window. No Round 1 prototype is expected.

References and source notes

Sources were checked during preparation. Citations in the report refer to the numbered list below. Statistics reflect each source's own definitions and sampling; links are direct and clickable. Research establishes context, not BrandLens performance.

[1] Inkloom AI / Code Axis Hackathon. Official brand-governance problem statement and Round 1 submission instructions. Team-provided brief image and organiser message (no public URL).

[2] ASCI. Annual Complaints Report 2025–26. Published 2026; reporting period Apr 2025–Mar 2026. Open source

[3] Central Consumer Protection Authority. Guidelines for Prevention of Misleading Advertisements and Endorsements for Misleading Advertisements, 2022. Open source

[4] McKinsey & Company. The state of AI in 2025: Agents, innovation, and transformation. Survey conducted Jun–Jul 2025; published Nov 2025. Open source

[5] WebAIM. The WebAIM Million 2026. Automated analysis of 1,000,000 homepages, Feb 2026; published Mar 2026. Open source

[6] Marq / Lucidpress. State of Brand Consistency (2021) infographic. Vendor survey; interpret as directional, self-reported evidence. Open source

[7] Acrolinx / Markup AI. Enterprise guidance product information (vendor claims; not independently tested). Open source

[8] Frontify Help Center. Brand Assistant. Current product help article; feature descriptions are vendor documentation. Open source

[9] Lewis, P. et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020. Open source

[10] Google AI for Developers. Gemini API structured output and pricing documentation. Features, rate limits and terms may change; synthetic data recommended for free-tier prototyping. Open source

[11] W3C. Web Content Accessibility Guidelines (WCAG) 2.2. Open source

[12] Technology documentation: Next.js, FastAPI, PostgreSQL and pgvector. Open source

[13] Google AI for Developers. Gemini API pricing and rate limits; verify current rates and free-tier data terms at build time. Open source

Research boundaries: no product tests, team interviews or surveys were represented as complete. Competitor descriptions are based on public vendor pages and are indicative, not an exhaustive competitive audit. Risk weights, outcomes and product performance claims remain proposed and unvalidated.